Fourth Section.

The Elliptic Functions.

§ 41. Connection of the ϑ-functions with the elliptic integrals.

By § 21 there exist, between the squares of the four ϑ-functions, two mutually independent linear equations, and one can therefore express two of these squares by means of the two others, or also express all four by two independent variables ξ, η. Doing the latter, let $\xi_1, \eta_1; \xi_2, \eta_2; \xi_3, \eta_3; \xi_4, \eta_4$ denote constants, and put, in connection with the notation of Vol. I, § 67,

$$\begin{aligned}
\vartheta_{01}^2(u) &= \xi\eta_1 - \eta\xi_1 = (\xi\eta_1),\\
\vartheta_{11}^2(u) &= \xi\eta_2 - \eta\xi_2 = (\xi\eta_2),\\
\vartheta_{10}^2(u) &= \xi\eta_3 - \eta\xi_3 = (\xi\eta_3),\\
\vartheta_{00}^2(u) &= \xi\eta_4 - \eta\xi_4 = (\xi\eta_4).
\end{aligned} \tag{1}$$

Between the constants ξ_i, η_i and the values $\vartheta_{01}, \vartheta_{10}, \vartheta_{00}$ there are four relations, which may be derived from the equations (13) of § 21. With the notation (1) these equations may be written in the form

$$\begin{aligned}
(\xi\eta_3)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{10}^2 - (\xi\eta_2)\vartheta_{00}^2,\\
(\xi\eta_4)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{00}^2 - (\xi\eta_2)\vartheta_{10}^2.
\end{aligned} \tag{2}$$

If in (2) one puts $\xi, \eta = \xi_1, \eta_1$ and $= \xi_2, \eta_2$, these relations can be given the form

$$\begin{aligned}
(\xi_1\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2,\\
(\xi_2\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2,\\
(\xi_1\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2,\\
(\xi_2\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2,
\end{aligned} \tag{3}$$

to which one can still add, by putting in (2) $\xi, \eta = \xi_4, \eta_4$ and using (3) and § 21, (14),

$$(\xi_4\eta_3) = (\xi_2\eta_1). \tag{4}$$

From (3) and (4) it follows for the double ratios that

$$\frac{(\xi_2\eta_3)(\xi_1\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{5}$$

$$\frac{(\xi_1\eta_2)(\xi_3\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{01}^4}{\vartheta_{00}^4}. \tag{6}$$

If we now differentiate two of the equations (1), say the first two, we obtain

$$2\vartheta_{01}(u)\vartheta_{01}'(u)\,du = \eta_1\,d\xi - \xi_1\,d\eta = (\eta_1\,d\xi),$$

$$2\vartheta_{11}(u)\vartheta_{11}'(u)\,du = \eta_2\,d\xi - \xi_2\,d\eta = (\eta_2\,d\xi),$$

and from these, using (1),

$$\begin{aligned}
&2\vartheta_{01}(u)\vartheta_{11}(u)\left[\vartheta_{01}'(u)\vartheta_{11}(u) - \vartheta_{11}'(u)\vartheta_{01}(u)\right]\,du\\
&\quad= (\eta_1\,d\xi)\vartheta_{11}^2(u) - (\eta_2\,d\xi)\vartheta_{01}^2(u)\\
&\quad= (\eta_1\,d\xi)(\xi\eta_2) - (\eta_2\,d\xi)(\xi\eta_1) = (\xi_2\eta_1)(\xi\,d\eta).
\end{aligned}$$

One can obtain the last expression without calculation by regarding the preceding expression as a linear function of $d\xi$, $d\eta$, which vanishes for $d\xi : d\eta = \xi : \eta$ and is therefore divisible by $(\xi\, d\eta)$. The quotient $(\xi_2\eta_1)$ is obtained by putting $d\xi : d\eta = \xi_2 : \eta_2$. With the use of § 23, (6) one thus obtains

$$2\pi\vartheta_{01}^2\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{10}(u)\vartheta_{01}(u)\, du = (\xi_1\eta_2)(\xi\, d\eta). \tag{7}$$

If, according to (3), the first and last formula are used to introduce

$$(\xi_1\eta_2)\vartheta_{00}^2 = \sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\,\vartheta_{01}^2, \tag{8}$$

and if one substitutes for the ϑ-functions the expressions (1), then finally

$$2\pi\vartheta_{00}^2\, du = \frac{\sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\,(\xi\, d\eta)}{\sqrt{(\xi\eta_1)(\xi\eta_2)(\xi\eta_3)(\xi\eta_4)}}, \tag{9}$$

whereby du is represented as an elliptic differential of the first kind in homogeneous variables [§ 1, (5)].

By (1) the ratio $\xi : \eta$ is determined as a doubly periodic function of u. Likewise the ratios of the square roots

$$\sqrt{(\xi\eta_1)}, \qquad \sqrt{(\xi\eta_2)}, \qquad \sqrt{(\xi\eta_3)}, \qquad \sqrt{(\xi\eta_4)} \tag{10}$$

are defined as single-valued doubly periodic functions, and the sign of the square roots in (9) is thereby, and by (8), also determined uniquely.

If ω is the modulus of the ϑ-functions, then, as follows from (1) by making the four ϑ-functions vanish, the following values belong together:

$$\begin{array}{r|l} u = -\frac{1}{2}\omega & \xi : \eta = \xi_1 : \eta_1, \\ u = 0 & \xi : \eta = \xi_2 : \eta_2, \\ u = \frac{1}{2} & \xi : \eta = \xi_3 : \eta_3, \\ u = \frac{1+\omega}{2} & \xi : \eta = \xi_4 : \eta_4. \end{array} \tag{11}$$

§ 42. Jacobi's elliptic functions.

Since one can replace the variables ξ, η by two new variables by means of a linear substitution in which four coefficients are disposable, one can assign arbitrary values to four of the quantities ξ_i, η_i, or to three of their ratios, without impairing the generality.

We shall put

$$\xi_1 = 0, \qquad \eta_2 = 0, \qquad \xi_3 = \eta_3, \tag{1}$$

and further introduce $\varkappa^2, \varkappa'^2, \zeta$ by the equations

$$\xi_4 = \varkappa^2\eta_4, \qquad \varkappa'^2 = 1 - \varkappa^2, \qquad \frac{\eta}{\xi} = \zeta. \tag{2}$$

From § 41, (3), (5), (6), there results

$$\begin{gathered} \frac{\xi_2}{\eta_1} = -\frac{\vartheta_{10}^2}{\vartheta_{00}^2}, \qquad \frac{\eta_3}{\eta_1} = \frac{\vartheta_{10}^2}{\vartheta_{01}^2}, \qquad \frac{\eta_4}{\eta_1} = \frac{\vartheta_{00}^2}{\vartheta_{01}^2}, \\ \sqrt{\varkappa} = \frac{\vartheta_{10}}{\vartheta_{00}}, \qquad \sqrt{\varkappa'} = \frac{\vartheta_{01}}{\vartheta_{00}}. \end{gathered} \tag{3}$$

From (1) and (9) of § 41 one finds:

$$\begin{aligned} \frac{\vartheta_{00}\vartheta_{11}(u)}{\vartheta_{10}\vartheta_{01}(u)} &= \sqrt{\zeta}, \\ \frac{\vartheta_{01}\vartheta_{10}(u)}{\vartheta_{10}\vartheta_{01}(u)} &= \sqrt{1-\zeta}, \\ \frac{\vartheta_{01}\vartheta_{00}(u)}{\vartheta_{00}\vartheta_{01}(u)} &= \sqrt{1-\varkappa^2\zeta}, \end{aligned} \tag{4}$$

and

$$2\pi\vartheta_{00}^2 u = \int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}. \tag{5}$$

In this formula the path of integration and the meaning of the root signs are determined by the formulae (4) themselves when the transition from 0 to u in the u-plane is given. The path of integration in (5), however, may then also be changed, provided only that none of the singular points $\infty, 1, 1/\varkappa^2$ is crossed.

The equation (5) can be written in the following three forms:

$$\begin{aligned} d\sqrt{\zeta} &= \pi\vartheta_{00}^2\sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\; du,\\ d\sqrt{1-\zeta} &= -\pi\vartheta_{00}^2\sqrt{\zeta(1-\varkappa^2\zeta)}\; du,\\ d\sqrt{1-\varkappa^2\zeta} &= -\pi\vartheta_{00}^2\varkappa^2\sqrt{\zeta(1-\zeta)}\; du. \end{aligned} \tag{6}$$

We introduce a new variable v by the equation

$$\pi\vartheta_{00}^2 u = v. \tag{7}$$

Then the equations (5), (6) become

$$v = \frac{1}{2}\int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{8}$$

$$\begin{aligned} d\sqrt{\zeta} &= \sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\; dv,\\ d\sqrt{1-\zeta} &= -\sqrt{\zeta(1-\varkappa^2\zeta)}\; dv,\\ d\sqrt{1-\varkappa^2\zeta} &= -\varkappa^2\sqrt{\zeta(1-\zeta)}\; dv. \end{aligned} \tag{9}$$

We now regard the three quantities $\sqrt{\zeta} = x$, $\sqrt{1-\zeta} = y$, $\sqrt{1-\varkappa^2\zeta} = z$ as functions of the variable v, defined by (4) and (7). By (4) they are single-valued, doubly periodic functions of v, continuous apart from isolated points at which they become infinite. Following Jacobi, they are called the sinus amplitudinis, cosinus amplitudinis, and Δ amplitudinis of v, and are denoted by $\sin\operatorname{am} v, \cos\operatorname{am} v, \Delta\operatorname{am} v$. We shall use here the shorter notation of Gudermann, already mentioned in § 14, namely $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$, whence

$$\begin{aligned} \frac{\vartheta_{11}(u)}{\vartheta_{01}(u)} &= \sqrt{\varkappa}\,\operatorname{sn} v,\\ \frac{\vartheta_{10}(u)}{\vartheta_{01}(u)} &= \sqrt{\frac{\varkappa}{\varkappa'}}\,\operatorname{cn} v,\\ \frac{\vartheta_{00}(u)}{\vartheta_{01}(u)} &= \frac{1}{\sqrt{\varkappa'}}\,\operatorname{dn} v. \end{aligned} \tag{10}$$

These functions satisfy, by (9), the differential equations

$$\frac{dx}{dv} = yz, \qquad \frac{dy}{dv} = -zx, \qquad \frac{dz}{dv} = -\varkappa^2 xy, \tag{11}$$

and the auxiliary conditions

$$\operatorname{sn} 0 = 0, \qquad \operatorname{cn} 0 = 1, \qquad \operatorname{dn} 0 = 1. \tag{12}$$

The relations between them are

$$y^2 = 1 - x^2, \qquad z^2 = 1 - \varkappa^2 x^2. \tag{13}$$

From the fundamental properties of the ϑ-functions the first properties of the elliptic functions follow:

$$\operatorname{sn} v = -\operatorname{sn}(-v), \qquad \operatorname{cn} v = \operatorname{cn}(-v), \qquad \operatorname{dn} v = \operatorname{dn}(-v),$$

that is, $\operatorname{sn} v$ is an odd function, while $\operatorname{cn} v$ and $\operatorname{dn} v$ are even functions.

Put further

$$\pi\vartheta_{00}^2 = 2K, \qquad \pi\vartheta_{00}^2\omega = 2iK', \tag{14}$$

and consequently

$$\pi\vartheta_{01}^2 = 2\varkappa' K, \qquad \pi\vartheta_{10}^2 = 2\varkappa K, \qquad \omega = \frac{iK'}{K}. \tag{15}$$

Then v assumes the values $K, iK', K+iK'$ when $u = \frac{1}{2}, \frac{\omega}{2}, \frac{1+\omega}{2}$; and from (4), with regard to (3) and § 21, (10), one obtains

$$\begin{aligned}
\operatorname{sn} K &= 1, & \operatorname{sn}(K+iK') &= \frac{1}{\varkappa}, \\
\operatorname{cn} K &= 0, & \operatorname{cn}(K+iK') &= -\frac{i\varkappa'}{\varkappa}, \\
\operatorname{dn} K &= \varkappa', & \operatorname{dn}(K+iK') &= 0, \\
\operatorname{sn} iK', &\quad \operatorname{cn} iK', & \operatorname{dn} iK' &= \infty.
\end{aligned} \tag{16}$$

Now K, K' may be expressed by means of (8) by definite integrals. If v is displaced from a vertex of the parallelogram $0, K, K+iK', iK'$ to the following one along the periphery, and hence u along

$$0, \frac{1}{2}, \frac{1+\omega}{2}, \frac{\omega}{2},$$

one obtains

$$K = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{17}$$

$$iK' = \frac{1}{2}\int_1^{1/\varkappa^2} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{18}$$

$$-K = \frac{1}{2}\int_{1/\varkappa^2}^{\infty} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{19}$$

$$-iK' = \frac{1}{2}\int_{\infty}^{0} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}. \tag{20}$$

The values of $\sqrt{\zeta}, \sqrt{1-\zeta}, \sqrt{1-\varkappa^2\zeta}$ in these integrations are determined by the formulae (4). If, however, ω is purely imaginary, and consequently $\varkappa^2$ is a positive proper fraction and K, K' are real and positive, then the paths for v are parallel to the real and imaginary axes, and, with regard to the remarks at the end of § 25, the formulae (4) show the following:

in (17)	$\sqrt{\zeta},\ \sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (18)	$\sqrt{\zeta},\ i\sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (19)	$\sqrt{\zeta},\ i\sqrt{1-\zeta},\ i\sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (20)	$-i\sqrt{\zeta},\ \sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive.

Moreover $\sqrt{\zeta}$ has no maximum or minimum on any of the paths of integration. Hence K, K' can be defined by the following four real integrals with positive square root:

$$K = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}} = \frac{1}{2}\int_{1/\varkappa^2}^{\infty} \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\varkappa^2\zeta-1)}}, \tag{21}$$

$$K' = \frac{1}{2}\int_{-\infty}^{0} \frac{d\zeta}{\sqrt{-\zeta(1-\zeta)(1-\varkappa^2\zeta)}} = \frac{1}{2}\int_1^{1/\varkappa^2} \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(1-\varkappa^2\zeta)}}. \tag{22}$$

§ 43. The Jacobi functions $\Theta(v), H(v)$.

If the ϑ-functions of the variable u are regarded as functions of v, one obtains the Jacobi Θ-functions. We put

$$\vartheta_{01}\left(\frac{v}{2K}, \omega\right) = \Theta(v), \qquad \vartheta_{11}\left(\frac{v}{2K}, \omega\right) = H(v). \tag{1}$$

Then $\Theta(v), H(v)$ are to be denoted as functions $t(v, 2K, 2K')$, and by § 25 one has

$$\begin{aligned} \Theta(v) &= 1 - 2q\cos\frac{\pi v}{K} + 2q^4\cos\frac{2\pi v}{K} - 2q^9\cos\frac{3\pi v}{K} + \cdots, \\ H(v) &= 2q^{1/4}\sin\frac{\pi v}{2K} - 2q^{9/4}\sin\frac{3\pi v}{2K} + 2q^{25/4}\sin\frac{5\pi v}{2K} - \cdots, \\ q &= e^{-\pi K'/K}. \end{aligned} \tag{2}$$

By § 21, (8),

$$\begin{aligned} H(v) &= -ie^{-\frac{\pi K'}{4K} + \frac{\pi i v}{2K}}\Theta(v + iK'), \\ \Theta(v) &= -ie^{-\frac{\pi K'}{4K} + \frac{\pi i v}{2K}}H(v + iK'). \end{aligned} \tag{3}$$

Furthermore,

$$\vartheta_{00}\left(\frac{v}{2K}, \omega\right) = \Theta(v + K), \qquad \vartheta_{10}\left(\frac{v}{2K}, \omega\right) = H(v + K), \tag{4}$$

whence, by means of the formulae (3), (10), (15) of the preceding paragraph and (5), § 23,

$$\begin{gathered} \sqrt{\varkappa} = \frac{H(K)}{\Theta(K)}, \qquad \sqrt{\varkappa'} = \frac{\Theta(0)}{\Theta(K)}, \\ \Theta(K) = \sqrt{\frac{2K}{\pi}}, \qquad \Theta(0) = \sqrt{\frac{2\varkappa' K}{\pi}}, \qquad H(K) = \sqrt{\frac{2\varkappa K}{\pi}}, \\ H'(0) = \frac{\Theta(0)H(K)}{\Theta(K)} = \sqrt{\frac{2\varkappa\varkappa' K}{\pi}}. \end{gathered} \tag{5}$$

Finally the representation of the elliptic functions is

$$\frac{H(v)}{\Theta(v)} = \sqrt{\varkappa}\,\mathrm{sn}\, v, \qquad \frac{H(v+K)}{\Theta(v)} = \sqrt{\frac{\varkappa}{\varkappa'}}\,\mathrm{cn}\, v, \qquad \frac{\Theta(v+K)}{\Theta(v)} = \frac{1}{\sqrt{\varkappa'}}\,\mathrm{dn}\, v. \tag{6}$$

The functions Θ, H have, as follows easily from (3), the periodicity expressed by the following equations:

$$\Theta(v + 2nK) = \Theta(v), \qquad \Theta(v + 2niK') = (-1)^n e^{\frac{n^2\pi K'}{K} - \frac{\pi i n v}{K}}\Theta(v), \tag{7}$$

$$H(v + 2nK) = (-1)^n H(v), \qquad H(v + 2niK') = (-1)^n e^{\frac{n^2\pi K'}{K} - \frac{\pi i n v}{K}}H(v). \tag{8}$$

It is sometimes useful to combine the equations (3), (7), (8) as follows:

$$\Phi(v + niK') = i^n e^{\frac{n^2\pi i K'}{4K} - \frac{\pi i n v}{2K}}\Psi(v), \tag{9}$$

where, if n is even, Φ and Ψ are both equal to Θ or both equal to H, while if n is odd, $\Phi = \Theta$, $\Psi = H$, or $\Phi = H$, $\Psi = \Theta$, is to be put.

From § 22, (2), (4), the addition formulae result:

$$\begin{aligned} \Theta^2(0)\Theta(u+v)\Theta(u-v) &= \Theta^2(u)\Theta^2(v) - H^2(u)H^2(v), \\ \Theta^2(0)H(u+v)H(u-v) &= H^2(u)\Theta^2(v) - \Theta^2(u)H^2(v), \end{aligned} \tag{10}$$

for which one may also write

$$\begin{aligned} \Theta^2(0)\Theta(u+v)\Theta(u-v) &= \Theta^2(u)\Theta^2(v)(1 - \varkappa^2\,\mathrm{sn}^2\, u\,\mathrm{sn}^2\, v), \\ \Theta^2(0)H(u+v)H(u-v) &= \Theta^2(u)\Theta^2(v)\varkappa(\mathrm{sn}^2\, u - \mathrm{sn}^2\, v). \end{aligned} \tag{11}$$

We now pass to transferring the theorems on the ϑ-functions to the elliptic functions, and begin with the addition theorem.

§ 44. Addition theorem of the elliptic functions.

From the four ϑ-functions one can form altogether twelve quotients, which by § 42 can be expressed by elliptic functions. If these quotients are formed for the argument $u+v$, then by § 22 one can express them by ϑ-functions of u and of v separately, and hence also by the corresponding elliptic functions, always in such a way that the denominator contains only squares of ϑ and is therefore expressed rationally by $\operatorname{sn}^2 u, \operatorname{sn}^2 v$. Let us, in order to carry out an example, take formula (5), § 22, and divide it first by (1) and then by (4), changing the first time in (5) v into $-v$; then

$$\frac{\vartheta_{01}}{\vartheta_{00}^2}\frac{\vartheta_{10}\vartheta_{11}(u+v)}{\vartheta_{00}(u+v)} = \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)+\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{00}^2(u)\vartheta_{00}^2(v)+\vartheta_{11}^2(u)\vartheta_{11}^2(v)},$$

$$\frac{\vartheta_{10}}{\vartheta_{01}}\frac{\vartheta_{00}(u+v)}{\vartheta_{11}(u+v)} = \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)-\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{11}^2(u)\vartheta_{01}^2(v)-\vartheta_{01}^2(u)\vartheta_{11}^2(v)}.$$

If u, v are replaced by $u : 2K, v : 2K$, everything can be expressed by elliptic functions according to § 42, (10). One obtains

$$\frac{\operatorname{sn}(u+v)}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{sn} u \operatorname{cn} v + \operatorname{dn} v \operatorname{sn} v \operatorname{cn} u}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{1}$$

$$\frac{\operatorname{dn}(u+v)}{\operatorname{sn}(u+v)} = \frac{\operatorname{dn} u \operatorname{sn} u \operatorname{cn} v - \operatorname{dn} v \operatorname{sn} v \operatorname{cn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}. \tag{2}$$

In the same way, from the formulae (6), (3), (4); (7), (2), (4); (8), (2), (3); (9), (1), (2); (10), (1), (3), of § 22, one obtains

$$\frac{\operatorname{sn}(u+v)}{\operatorname{cn}(u+v)} = \frac{\operatorname{cn} u \operatorname{sn} u \operatorname{dn} v + \operatorname{cn} v \operatorname{sn} v \operatorname{dn} u}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{3}$$

$$\frac{\operatorname{cn}(u+v)}{\operatorname{sn}(u+v)} = \frac{\operatorname{cn} u \operatorname{sn} u \operatorname{dn} v - \operatorname{cn} v \operatorname{sn} v \operatorname{dn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \tag{4}$$

$$\operatorname{sn}(u+v) = \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v + \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{5}$$

$$\frac{1}{\operatorname{sn}(u+v)} = \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v - \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \tag{6}$$

$$\operatorname{cn}(u+v) = \frac{\operatorname{cn} u \operatorname{cn} v - \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{7}$$

$$\frac{1}{\operatorname{cn}(u+v)} = \frac{\operatorname{cn} u \operatorname{cn} v + \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{8}$$

$$\operatorname{dn}(u+v) = \frac{\operatorname{dn} u \operatorname{dn} v - \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{9}$$

$$\frac{1}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v + \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{10}$$

$$\frac{\operatorname{dn}(u+v)}{\operatorname{cn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v \operatorname{cn} u \operatorname{cn} v + \varkappa'^2 \operatorname{sn} u \operatorname{sn} v}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{11}$$

$$\frac{\operatorname{cn}(u+v)}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v \operatorname{cn} u \operatorname{cn} v - \varkappa'^2 \operatorname{sn} u \operatorname{sn} v}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \tag{12}$$

One can still derive many other formulae in the same way. Among them we record the following three, which follow by division of § 22, (4), (3), (1), by (2):

$$\operatorname{sn}(u+v)\operatorname{sn}(u-v) = \frac{\operatorname{sn}^2 u - \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{13}$$

$$\operatorname{cn}(u+v)\operatorname{cn}(u-v) = \frac{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{14}$$

$$\operatorname{dn}(u+v)\operatorname{dn}(u-v) = \frac{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \tag{15}$$

The most important among these formulae are (5), (7), (9); from them, although by rather lengthy calculations, the others can all be derived. For clarity we put them once more in order:

$$\begin{aligned}\operatorname{sn}(u+v) &= \frac{\operatorname{sn}u\operatorname{cn}v\operatorname{dn}v+\operatorname{sn}v\operatorname{cn}u\operatorname{dn}u}{1-\varkappa^2\operatorname{sn}^2u\operatorname{sn}^2v},\\ \operatorname{cn}(u+v) &= \frac{\operatorname{cn}u\operatorname{cn}v-\operatorname{sn}u\operatorname{sn}v\operatorname{dn}u\operatorname{dn}v}{1-\varkappa^2\operatorname{sn}^2u\operatorname{sn}^2v},\\ \operatorname{dn}(u+v) &= \frac{\operatorname{dn}u\operatorname{dn}v-\varkappa^2\operatorname{sn}u\operatorname{sn}v\operatorname{cn}u\operatorname{cn}v}{1-\varkappa^2\operatorname{sn}^2u\operatorname{sn}^2v}.\end{aligned}\tag{16}$$

By combining two of these equations at a time, one can give them, among others, the forms

$$\begin{aligned}\operatorname{cn}(u+v)\operatorname{dn}u\operatorname{dn}v-\operatorname{dn}(u+v)\operatorname{cn}u\operatorname{cn}v &= -\varkappa'^2\operatorname{sn}u\operatorname{sn}v,\\ \operatorname{dn}(u+v)\operatorname{sn}u\operatorname{cn}v-\operatorname{sn}(u+v)\operatorname{dn}u &= -\operatorname{cn}u\operatorname{sn}v,\\ \operatorname{sn}(u+v)\operatorname{cn}u-\operatorname{cn}(u+v)\operatorname{sn}u\operatorname{dn}v &= \operatorname{dn}u\operatorname{sn}v,\\ \operatorname{cn}(u+v)\operatorname{cn}u+\operatorname{sn}(u+v)\operatorname{sn}u\operatorname{dn}v &= \operatorname{cn}v.\end{aligned}\tag{17}$$

We first apply the formulae (16) to establishing the periodicity of the elliptic functions, which of course can also be derived from § 21. If in (16) one puts $v=\pm K$, $v=K+iK'$, then by § 42, (16),

$$\operatorname{sn}(u\pm K)=\pm\frac{\operatorname{cn}u}{\operatorname{dn}u},\qquad \operatorname{cn}(u\pm K)=\mp\frac{\varkappa'\operatorname{sn}u}{\operatorname{dn}u},\qquad \operatorname{dn}(u\pm K)=\frac{\varkappa'}{\operatorname{dn}u},\tag{18}$$

$$\operatorname{sn}(u+K+iK')=\frac{1}{\varkappa}\frac{\operatorname{dn}u}{\operatorname{cn}u},\qquad \operatorname{cn}(u+K+iK')=\frac{i\varkappa'}{\varkappa}\frac{1}{\operatorname{cn}u},\qquad \operatorname{dn}(u+K+iK')=\frac{i\varkappa'\operatorname{sn}u}{\operatorname{cn}u}.\tag{19}$$

If in (19) u is changed into $u-K$, then

$$\operatorname{sn}(u+iK')=\frac{1}{\varkappa}\frac{1}{\operatorname{sn}u},\qquad \operatorname{cn}(u+iK')=-\frac{i}{\varkappa}\frac{\operatorname{dn}u}{\operatorname{sn}u},\qquad \operatorname{dn}(u+iK')=-i\frac{\operatorname{cn}u}{\operatorname{sn}u}.\tag{20}$$

The formulae (20) show that the three functions $\operatorname{sn}u,\operatorname{cn}u,\operatorname{dn}u$ become infinite for $u=iK'$. Forming the quotients of two at a time gives

$$\lim\frac{\operatorname{sn}u}{\operatorname{cn}u}=i,\qquad \lim\frac{\operatorname{sn}u}{\operatorname{dn}u}=i\qquad \text{for } u=iK'.\tag{21}$$

With the help of the relations (18), (19), (20), one can derive from each addition formula three others, by replacing u by $u+K$, $u+K+iK'$, $u+iK'$, and hence making the substitutions collected in the following table:

$\operatorname{sn}u,$	$\operatorname{cn}u,$	$\operatorname{dn}u,$
$\frac{\operatorname{cn}u}{\operatorname{dn}u},$	$-\frac{\varkappa'\operatorname{sn}u}{\operatorname{dn}u},$	$\frac{\varkappa'}{\operatorname{dn}u},$
$\frac{1}{\varkappa}\frac{\operatorname{dn}u}{\operatorname{cn}u},$	$\frac{i\varkappa'}{\varkappa\operatorname{cn}u},$	$\frac{i\varkappa'\operatorname{sn}u}{\operatorname{cn}u},$
$\frac{1}{\varkappa\operatorname{sn}u},$	$-\frac{i\operatorname{dn}u}{\varkappa\operatorname{sn}u},$	$-\frac{i\operatorname{cn}u}{\operatorname{sn}u}.$

The same substitutions are to be applied also to $\operatorname{sn}(u\pm v)$, $\operatorname{cn}(u\pm v)$, $\operatorname{dn}(u\pm v)$. Thus, for example, the twelve formulae (1) to (12) arise from the three formulae (16).

If these substitutions are applied to the relations (18), (19), (20) themselves, one obtains

$$\begin{aligned}\operatorname{sn}(u+2K)&=-\operatorname{sn}u, & \operatorname{sn}(u+2iK')&=\operatorname{sn}u,\\ \operatorname{cn}(u+2K)&=-\operatorname{cn}u, & \operatorname{cn}(u+2iK')&=-\operatorname{cn}u,\\ \operatorname{dn}(u+2K)&=\operatorname{dn}u, & \operatorname{dn}(u+2iK')&=-\operatorname{dn}u.\end{aligned}\tag{22}$$

The common periods of these three functions are therefore $4K, 4iK'$; but in addition $\operatorname{sn}u$ has the period $2iK'$, $\operatorname{dn}u$ the period $2K$, and $\operatorname{cn}u$ the period $2K+2iK'$.

We now give the theorem on Θ-functions (§ 21) only another expression, by stating the following theorem: every doubly periodic function of v with the periods $2K, 2iK'$ which everywhere has the character of a rational function is, if it is an even function, a rational function of $\operatorname{sn}^2 v$, and if it is an odd function, the product of $\operatorname{sn} v \operatorname{cn} v \operatorname{dn} v$ with a rational function of $\operatorname{sn}^2 v$.

Thus every theorem which relates to homogeneous functions of degree zero of ϑ-functions can be transformed into a theorem on elliptic functions; and from this one again obtains corresponding theorems on elliptic integrals. One sees at once that the addition formulae (16) are no others than the formulae (17) of § 13 in the first section. In the same way the transformation theory of the ϑ-functions gives us a solution of Jacobi's transformation problem for the elliptic integrals, as we presented it in § 8.

We shall prove this for the two principal transformations of the second order, which in § 32 as well as in § 9 we called the Gaussian and Landen transformations. By § 32, for the Gaussian transformation,

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right) \vartheta_{11}\left(u, \frac{\omega}{2}\right)}{\vartheta_{00}\left(0, \frac{\omega}{2}\right) \vartheta_{01}\left(u, \frac{\omega}{2}\right)} = \frac{2\vartheta_{01}(u)\vartheta_{11}(u)}{\vartheta_{01}^2(u) + \vartheta_{11}^2(u)}, \tag{23}$$

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right)^2}{\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2} = \frac{2\vartheta_{10}\vartheta_{00}}{\vartheta_{00}^2 + \vartheta_{10}^2} = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \tag{24}$$

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2 = \vartheta_{00}^2 + \vartheta_{10}^2 = (1+\varkappa)\vartheta_{00}^2. \tag{25}$$

According to § 42 we therefore obtain a relation between elliptic functions with two different moduli. If λ, L, L' arise from $\varkappa, K, K'$ by the interchange $(\omega, \omega/2)$, then by (23), (24), (25)

$$\lambda = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \qquad L = (1+\varkappa)K, \qquad L' = (1+\varkappa)K', \tag{26}$$

$$\operatorname{sn}[(1+\varkappa)v, \lambda] = \frac{(1+\varkappa)\operatorname{sn} v}{1+\varkappa \operatorname{sn}^2 v}, \tag{27}$$

where, for clarity, the modulus is included under the function sign sn as a second argument.

For the Landen transformation,

$$\frac{\vartheta_{11}(2u, 2\omega)}{\vartheta_{01}(2u, 2\omega)} = \frac{\vartheta_{10}(u)\vartheta_{11}(u)}{\vartheta_{00}(u)\vartheta_{01}(u)},$$

$$\frac{\vartheta_{01}(0, 2\omega)^2}{\vartheta_{00}(0, 2\omega)^2} = \frac{2\vartheta_{00}\vartheta_{01}}{\vartheta_{00}^2 + \vartheta_{01}^2} = \frac{2\sqrt{\varkappa'}}{1+\varkappa'},$$

$$2\vartheta_{00}^2(0, 2\omega) = (1+\varkappa')\vartheta_{00}^2.$$

It follows, when λ, λ', L, L' arise from $\varkappa, \varkappa', K, K'$ by the interchange $(\omega, 2\omega)$, that

$$\lambda' = \frac{2\sqrt{\varkappa'}}{1+\varkappa'}, \qquad \sqrt{\lambda} = \frac{\varkappa}{1+\varkappa'}, \qquad \lambda = \frac{1-\varkappa'}{1+\varkappa'}, \tag{28}$$

$$2L = (1+\varkappa')K, \qquad L' = (1+\varkappa')K',$$

$$\operatorname{sn}[(1+\varkappa')v, \lambda] = \frac{(1+\varkappa')\operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \tag{29}$$

and in (26) to (29), according to § 42, one recognizes the formulae (2) and (4) of § 9.

§ 45. Linear transformation of the elliptic functions.

The action of linear transformation on the elliptic functions is best surveyed from their representation by the σ-functions. From (4) or (10) of § 42 and the formulae (4), (6) of § 37 one obtains

$$\frac{\omega_1}{2K}\operatorname{sn}\frac{2Ku}{\omega_1}=\frac{\sigma(u)}{\sigma_{01}(u)},\qquad \operatorname{cn}\frac{2Ku}{\omega_1}=\frac{\sigma_{10}(u)}{\sigma_{01}(u)},\qquad \operatorname{dn}\frac{2Ku}{\omega_1}=\frac{\sigma_{00}(u)}{\sigma_{01}(u)},\tag{1}$$

or also, using the homogeneity of the σ-function [§ 37, (8)],

$$\operatorname{sn}v=\frac{\sigma(v,2K,2iK')}{\sigma_{01}(v,2K,2iK')},\qquad \operatorname{cn}v=\frac{\sigma_{10}(v,2K,2iK')}{\sigma_{01}(v,2K,2iK')},\qquad \operatorname{dn}v=\frac{\sigma_{00}(v,2K,2iK')}{\sigma_{01}(v,2K,2iK')}.\tag{2}$$

The σ-functions occurring on the right side of (2) depend only on the two variables v, ω, and it is first a matter of determining the change of these functions under application of the fundamental transformations

$$(\omega,\omega+1),\qquad \left(\omega,-\frac{1}{\omega}\right).$$

Because

$$\pi\vartheta_{00}^2=2K,\qquad 2iK'=2\omega K,$$

one obtains from § 31, (6), (11), the corresponding changes:

$$\begin{array}{ccccc}
\omega, & 2K, & 2iK', & \varkappa, & \varkappa',\\
\omega+1, & 2\varkappa' K, & 2\varkappa'(K+iK'), & \frac{i\varkappa}{\varkappa'}, & \frac{1}{\varkappa'},\\
-1/\omega, & 2K', & 2iK, & \varkappa', & \varkappa.
\end{array}$$

Put temporarily

$$f(v,\omega)=\sigma(v,2K,2iK').$$

Then

$$\begin{aligned}
f(v,\omega+1)&=\sigma[v,2\varkappa' K,2\varkappa'(K+iK')]\\
&=\sigma(v,2\varkappa' K,2\varkappa' iK')\qquad\text{[by § 35, (7)]}\\
&=\varkappa'\sigma\left(\frac{v}{\varkappa'},2K,2iK'\right)\qquad\text{[by § 37, (8)]},\\
f\left(v,-\frac{1}{\omega}\right)&=\sigma(v,2K',2iK)\\
&=\sigma(v,-2iK,2K')\qquad\text{[by § 35, (7)]}\\
&=-i\sigma(iv,2K,2iK')\qquad\text{[by § 37, (8)]}.
\end{aligned}$$

Doing the corresponding calculation for the other three σ-functions and taking § 36, (7), into account, one obtains the following corresponding substitutions:

$$\begin{array}{ccccc}
\omega, & \sigma(v), & \sigma_{00}(v), & \sigma_{01}(v), & \sigma_{10}(v),\\
\omega+1, & \varkappa'\sigma\left(\frac{v}{\varkappa'}\right), & \sigma_{01}\left(\frac{v}{\varkappa'}\right), & \sigma_{00}\left(\frac{v}{\varkappa'}\right), & \sigma_{10}\left(\frac{v}{\varkappa'}\right),\\
-1/\omega, & -i\sigma(iv), & \sigma_{00}(iv), & \sigma_{10}(iv), & \sigma_{01}(iv),
\end{array}\tag{3}$$

where the periods are $2K, 2iK'$. From this, by (2), follow the first two linear transformations of the elliptic functions:

$$\begin{aligned}
\operatorname{sn}\left(\varkappa' v,\frac{i\varkappa}{\varkappa'}\right)&=\varkappa'\frac{\operatorname{sn}v}{\operatorname{dn}v},\\
\operatorname{cn}\left(\varkappa' v,\frac{i\varkappa}{\varkappa'}\right)&=\frac{\operatorname{cn}v}{\operatorname{dn}v},\\
\operatorname{dn}\left(\varkappa' v,\frac{i\varkappa}{\varkappa'}\right)&=\frac{1}{\operatorname{dn}v},
\end{aligned}\tag{4}$$

$$
\begin{aligned}
\operatorname{sn}(iv, \varkappa') &= i\frac{\operatorname{sn} v}{\operatorname{cn} v},\\
\operatorname{cn}(iv, \varkappa') &= \frac{1}{\operatorname{cn} v},\\
\operatorname{dn}(iv, \varkappa') &= \frac{\operatorname{dn} v}{\operatorname{cn} v}.
\end{aligned} \tag{5}
$$

The first formula (5) shows by § 44, (21), that $\operatorname{sn}(v, \varkappa') = 1$ when $v = K'$; hence by § 42, (17), the second frequently used representation for K' is obtained:

$$
K' = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa'^2\zeta)}}. \tag{6}
$$

From (4), (5) one obtains the remaining cases of linear transformation by repeated application. Put in (4) $iv, \varkappa'$ in place of $v, \varkappa$ and apply (5); then

$$
\begin{aligned}
\operatorname{sn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= i\varkappa\frac{\operatorname{sn} v}{\operatorname{dn} v},\\
\operatorname{cn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{1}{\operatorname{dn} v},\\
\operatorname{dn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{\operatorname{cn} v}{\operatorname{dn} v}.
\end{aligned} \tag{7}
$$

Conversely, replace in (5) $v, \varkappa, \varkappa'$ by $\varkappa' v, \frac{i\varkappa}{\varkappa'}, \frac{1}{\varkappa'}$, and apply (4). Then

$$
\begin{aligned}
\operatorname{sn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= i\varkappa'\frac{\operatorname{sn} v}{\operatorname{cn} v},\\
\operatorname{cn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{\operatorname{dn} v}{\operatorname{cn} v},\\
\operatorname{dn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{1}{\operatorname{cn} v}.
\end{aligned} \tag{8}
$$

If in this one again replaces v by iv and $\varkappa, \varkappa'$ by $\varkappa', \varkappa$, and applies (5), one finds

$$
\begin{aligned}
\operatorname{sn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \varkappa \operatorname{sn} v,\\
\operatorname{cn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{dn} v,\\
\operatorname{dn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{cn} v,
\end{aligned} \tag{9}
$$

with which the six classes of linear transformations are exhausted, since more frequent repetition gives occasion for no new formulae.